

3 · Experiments

Datasets, baselines, and key results.

Setup

QM9 (small organics, up to 9 atoms) and ZINC-250k (drug-like). 500 diffusion steps; classifier-free guidance, $\lambda = 3$ on QM9, $\lambda = 2$ on ZINC. Two tasks. Targeting: 100 property values $\times$ 10 molecules each, scored by MAE. Optimization (Jin et al.): corrupt for 100 steps, denoise toward the target, keep the best candidate above a Tanimoto threshold.

Baselines

FreeGress and DiGress for targeting; JT-VAE, CG-VAE, GCPN for optimization. All operate on a fixed graph size. GrIDDD is the only model that can insert and delete atoms while sampling.

Targeting accuracy

GrIDDD lowers MAE on both targeting benchmarks. QM9 dipole moment μ : 0.66 vs 0.74 (FreeGress). ZINC molecular weight: 4.89 vs 8.96, roughly half the error of the strongest baseline, with no target size supplied to the model at inference.

Optimization, OOD, ablation

On graph-property optimization, QED success reaches 45.1% against 9.4% for GCPN, and LogP improvement is 2.70. Out of distribution, where models are trained up to 14 atoms, GrIDDD still produces 35% valid molecules at size 15; DiGress collapses to near zero. Removing insertion and deletion takes QED success from 45.1% to 33.8%, so size control, not the diffusion backbone alone, is what produces the gains.

(a) ZINC-250k targeting

Table 2: Results of property targeting on the ZINC-250k dataset.						
Method	LogP		QED		MW	
	MAE ↓	Val. ↑	MAE ↓	Val. ↑	MAE ↓	Val. ↑
Unconditional	1.52 ± 0.12	86.1 %	0.15 ± 0.01	86.1 %	74.16 ± 6.71	86.1 %
DiGress	0.74 ± 0.08	74.6 %	0.15 ± 0.01	85.1 %	20.92 ± 2.90	40.4 %
FreeGress	0.17 ± 0.01	84.9 %	0.04 ± 0.01	84.9 %	8.96 ± 1.93	79.7 %
GrIDDD	0.19 ± 0.02	87.9 %	0.04 ± 0.00	87.2 %	4.89 ± 0.57	84.2%

(b) OOD validity

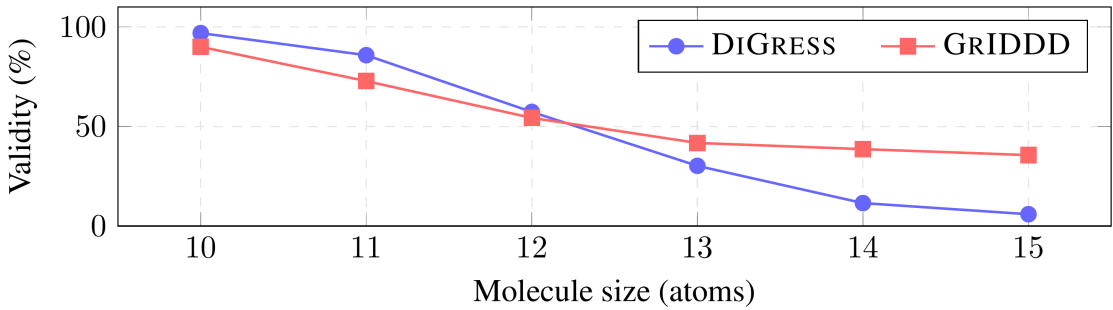


Figure 3: Validity results of out-of-distribution sampling on QM9.

(c) Graph-property optimization

Table 3: Benchmark on graph property optimization.								
Method	LogP (sim ≥ 0.4)		LogP (sim ≥ 0.6)		QED (sim ≥ 0.4)		DRD2 (sim ≥ 0.4)	
	Improv. ↑	Div.↑	Improv. ↑	Div.↑	Succ.↑	Div.↑	Succ.↑	Div.↑
JT-VAE	1.03 ± 1.39	-	0.28 ± 0.79	-	8.8 %	-	3.4 %	-
CG-VAE	0.61 ± 1.09	-	0.25 ± 0.74	-	4.8 %	-	-	-
GCPN	2.49 ± 1.30	-	0.79 ± 0.63	-	9.4 %	0.216	4.4 %	0.152
GrIDDD	2.70 ± 0.94	0.482	1.33 ± 0.61	0.280	45.1 %	0.283	5.0 %	0.121

Quantitative results: (a) ZINC-250k targeting; (b) out-of-distribution validity vs molecule size; (c) graph-property optimization (Ninniri et al., 2025).